Seminar 8

19.11.2025

(5.) 4. Choose an interpretation (arbitrary) with a finite domain (2 elements) for the formula U_4 and prove that it is a model of U_4.

$$U_4 = (\exists x)(A(x) \to B(x)) \leftrightarrow ((\forall)x)A(x) \to (\exists x)B(x).$$

A, B → unary predicate symbols.

~~if U ≡ F then~~

U

Interpretation:

$I = \langle D, m \rangle$

D → domain of interpretation

m → function

$I = \langle D, m \rangle$

$D = \{4, 5\}$

$m(A): D \to \{T, F\}$, $m(A)(x) =$ "x is a prime".

$m(B): D \to \{T, F\}$, $m(B)(x) =$ "x is odd".

$$v^I(U_4) = v^I((\exists x)(A(x) \to B(x)) \leftrightarrow v^I((\forall)xA(x) \to (\exists x)B(x))$$

$$= v^I((\exists x)(A(x) \to B(x))) \leftrightarrow (v^I((\forall x)A(x)) \to v^I((\exists x)B(x)))$$

$$= (A(4) \to B(4)) \vee (A(5) \to B(5)) \leftrightarrow (A(4) \wedge A(5)) \to (B(4) \vee B(5))$$

$$\boxed{(\forall x)_{x \in \{a,b\}} U(x) \equiv U(a) \wedge U(b)}$$

$$\boxed{(\exists x)_{x \in \{a,b\}} U(x) \equiv U(a) \vee U(b)}$$

$$= (F \to F) \vee (T \to T) \leftrightarrow (F \wedge T) \to (F \vee T)$$

$$= T \vee T \leftrightarrow F \to T$$

$$= T \leftrightarrow T$$

$$= T$$

⇒ I model of U_4.

6.4 $U_4 = (\exists x)P(x) \land (\exists x)Q(x) \to (\exists x)(P(x) \land Q(x))$

cannot be decomposed anymore bc. it has a main quantifier.

Find a model and an anti-model for U_4.

An element such that ..

$I_1 = \langle D, m \rangle$, $D = \mathbb{N}$.

$m(P)(x) = $ "$x < 0$"

$m(Q)(x) = $ "x even". ← it does not matter who is Q.

$$v^I(U_4) = v^I((\exists x)P(x) \land v^I((\exists x)Q(x)) \to v^I((\exists x)(P(x) \land Q(x)))$$

$$= (\exists x)_{x \in \mathbb{N}} \text{"}x<0\text{"} \land (\exists x)_{x \in \mathbb{N}} \text{"}x \text{ even"} \to$$

$$\to (\exists x)_{x \in \mathbb{N}} (\text{"}x<0\text{"} \land \text{"}x \text{ even"})$$

$$= F \land T \to F$$

$$= F \to F$$

$$= T.$$

I_1 → model

(implication is F when the first T, the second F.)

if $U \equiv F$ then $U \to V \equiv T$

$\underbrace{U}_{T} \to \underbrace{V}_{F} \equiv F$

$I_2 = \langle D, m \rangle$

$D = \mathbb{N}$. → see examples in the lecture → the set of all lines I, II → prime, composite) → odd, even? → things that (concepts) exclude each other.

$m(P)(x) = $ "x even"

$m(Q)(x) = $ "x odd".

$$v^I(U_4) = v^I((\exists x)P(x) \land v^I((\exists x)Q(x))$$

$$\to v^I((\exists x)(P(x) \land Q(x)))$$

$$= (\exists x)_{x \in \mathbb{N}} \text{"}x \text{ even"} \land (\exists x)_{x \in \mathbb{N}} \text{"}x \text{ odd"}$$

$$\to (\exists x)_{x \in \mathbb{N}} (\text{"}x \text{ even"} \land \text{"}x \text{ odd"}) = T \land T \to F = T \to F = F.$$

⇒ I_2 antimodel of U_4.

7. HW: 2, 4 → Resolution in propositional logic

Decomposition rules

- conjunctive formulas – α rules

$$A \land B \;\to\; A,\ B \qquad \neg(A \lor B) \equiv \neg A \land \neg B \;\to\; \neg A,\ \neg B \qquad \neg(A \to B) \equiv A \land \neg B \;\to\; A,\ \neg B$$

- disjunctive formulas – β rules

$$A \lor B \;\to\; A \mid B \qquad \neg(A \land B) \equiv \neg A \lor \neg B \;\to\; \neg A \mid \neg B \qquad A \to B \equiv \neg A \lor B \;\to\; \neg A \mid B$$

1.4 Using the semantic tableaux method, decide if it's consistent, inconsistent, valid.

If consistent $\Rightarrow$ models.

$U_4 = (q \lor r \to p) \to (p \to r) \land q$

1. U consistent if it has an open semantic tableau. The open branches provide the models of U.

2. U inconsistent if it has a closed semantic tableau.

3. $\models U$ iff $\neg U$ has a closed semantic tableau.

   $\Updownarrow$

   $\neg U$ inconsistent $\Rightarrow U$ valid (PROOF BY CONTRADICTION)

4. $U_1, \ldots, U_n \models V$ iff $U_1 \land \ldots \land U_n \land \neg V$ has a closed semantic tableau.

5. The anti-models of U are the models of $\neg U$ and are provided by the open branches of the sem. tableau of $\neg U$.

$U_4 = (q \lor r \to p) \to (p \to r) \land q \quad (1) \checkmark$

– β rule for (1)

$\neg(q \lor r \to p)$ (2) ✓ $\qquad (p \to r) \land q$ (3)

α rule for (2) $\qquad$ α rule for (3)

$q \lor r$ (4) ✓ $\qquad p \to r$ (5) ✓

$\neg p$ (5) $\qquad q$

β rule for (4) $\qquad$ β-rule for (5)

$q \quad r \qquad \neg p \quad r$

$\odot \quad \odot \qquad \odot \quad \odot$

we constantly search for opposite literals.

we found an OPEN BRANCH.

$\Rightarrow$ it is consistent formula,

but we will continue anyways.

toate valide $\Rightarrow$ consistent.

$\Rightarrow U_4$ has an open semantic tableau, with 4 open branches, so it is consistent.

$\Rightarrow$ DNF$(U_m) = (\neg p \land q) \lor (\neg p \land r)$ ~~$\lor (\neg p \land r) \lor$~~ $\lor (q \land \neg p) \lor (q \land r)$

in a graphical way

Cube $\neg p \lor q \equiv T$ provides:

$i_1, i_2 : \{p, q, r\} \to \{T, F\}$.

$i_1(p) = F, \; i_1(q) = T, \; i_1(r) = T$.

$i_2(p) = F \quad i_2(q) = T \quad i_2(r) = F$

$i_1, i_2(U_4)$ T when i_1, i_2 models of U_4.

Cube $\neg p \land r$ provides..

continue with the other models.

2.4 Prove that U_4 tautology.

$$\boxed{\begin{array}{l} A \leftrightarrow B \equiv (A \to B) \wedge (B \to A) \\ \text{equiv} = \text{conj. of 2 implications} \end{array}}$$

~~$U_4 = p \vee$~~ $U = A \leftrightarrow B$

$U_1 .\ A \to B$

$U_2 .\ B \to A$

$\models U$ iff $\models U_1$ and $\models U_2$.

$$U_4 = \underbrace{p \vee (q \leftrightarrow r)}_{A} \leftrightarrow \underbrace{((p \vee q) \leftrightarrow (p \vee r))}_{B}$$

$V = A \to B$

$= p \vee (q \leftrightarrow r) \leftrightarrow ((p \vee q) \leftrightarrow (p \vee r))$

$\neg V = \neg((p \vee (q \leftrightarrow r)) \to ((p \vee q) \leftrightarrow (p \vee r)))$ (1) ✓

| α rule (1)

$p \vee (q \leftrightarrow r)$ (2) ✓

$\neg((p \vee q \to p \vee r) \wedge (p \vee r \to p \vee q)) \equiv \neg((p \vee q) \leftrightarrow (p \vee r))$

β ~~meth~~ rule (2)

p — β rule (3) — $q \leftrightarrow r \equiv (q \to r) \wedge (r \to q)$ (4)

$\neg(p \vee q \to p \vee r)$ (7)

| α rule (7)

$p \vee q$ (9)

$\neg(p \vee r)$ (10)

| α rule (10)

$\neg p$

$\neg r$

ⓧ

$\neg(p \vee r \to p \vee q)$ (8)

α rule (8)

$p \vee r$ (11)

$\neg(p \vee q)$ (12)

| α rule (12)

$\neg p$

$\neg q$

ⓧ

| α rule (4)

$q \to r$ (6)

$r \to q$ (5)

β rule (5)

$\neg r$ | q

β rule (6)

$\neg q$ | r | $\neg q$ | r

ⓧ | ⓧ | ⓧ | ⓧ

=) $\neg V$ has a closed semantic table with 6 closed branches. =) V tautology.

3) Using the semantic tableau method, decide:

~~$p\to q$, $r\to t$, $p\wedge r$~~

$$\underbrace{p\to q}_{U_1}, \underbrace{r\to t}_{U_2}, \underbrace{p\wedge r}_{U_3} \overset{?}{\models} \underbrace{q\wedge t}_{V}$$

$U_1 \wedge U_2 \wedge U_3 \wedge \neg V$ (1)

| α rule (1)

U_1: $p\to q$ (2) ✓

U_2: $r\to t$ (3) ✓

U_3: $p\wedge r$ (4) ✓

$\neg V$: $\neg q \vee \neg t$

| α rule for U_3 (4)

p

r

β for (2)

$\neg p$ ✗ closed with p q

β for (3)

r t

β for (5)

$\neg q$ ✗ $\neg t$ ✗

$U_1 \wedge U_2 \wedge U_3 \wedge \neg V$ has a closed sem. table with 4 closed branches

$(p, \neg p)$

$(r, \neg r)$, $(q, \neg q)$, $(t, \neg t)$ so $U_1, U_2, U_3 \models V$.